

Signed Rank Test - Lab Activity

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Dealing with Discrete Distributions

Figure out how you would find the p-value or the appropriate critical value (to define the critical or rejection region) for a signed rank test procedure as specified in the problems below. Compute it using R! Remember to be careful due to the discrete nature of these distributions. Think about: do I need to add 1? Subtract 1? Leave the value alone? We recommend sketches and writing down the desired probabilities of interest before trying to compute them.

1. $n=15$, $T+ = 85$, upper-tailed test. Compute the p-value.

SOLUTION:

2. $n=15$, upper-tailed test. Find the critical value that comes closest to (but without going over) an $\alpha=0.03$ level test.

SOLUTION:

3. $n=15$, $T+ = 24$, two-tailed test. Compute the p-value.

SOLUTION:

4. $n=15$, two-tailed test. Find the critical value that comes closest to (but without going over) an $\alpha=0.08$ level test. Remember the distribution is symmetric around the mean in the case of no ties, which can help your computations, and $(Max(T)+ = n(n+1)/2)$.

SOLUTION:

Practice Problems in R

Complete the following problems based on the provided data. Bear in mind you are allowed to perform parametric procedures if appropriate conditions check out, and this data set has a large sample size.

```
abalone <- read.table("https://awagaman.people.amherst.edu/nonparametricstats/abalone.txt",  
                     h = T, sep = ",")
```

This data set was stored as a comma-delimited text file, so we read it in as a table, and had to use the sep option to tell it about the commas. We will usually see read.table or read.csv commands to load data.

This data set contains information on 4177 abalones (sea snails). The variables included are:

- Sex - M, F, and I (infant)
- Length - mm - Longest shell measurement
- Diameter - mm - perpendicular to length
- Height - mm - with meat in shell
- Whole weight - grams - whole abalone
- Shucked weight - grams - weight of meat
- Viscera weight - grams - gut weight
- Shell weight - grams - after being dried
- Rings - number of rings in the shell - 1.5+rings gives the age in years

Note that the descriptions do not exactly match the variable names - whoever entered the data shortened the names somewhat. To see the names of variables in the data set, use:

```
names(abalone)
```

```
## [1] "Sex"      "Length"   "Diameter" "Height"   "Wholewt"  "Shuckedwt"  
## [7] "Viscerawt" "Shellwt"  "Rings"
```

Now, let's try to address some questions that researchers have about the sea snails. For the two questions below, perform both an appropriate parametric and nonparametric procedure and compare the results.

Question 1

Is 0.3 a reasonable value for a randomly selected abalone's diameter?

Remember to think through conditions for applying your chosen statistical procedures.

SOLUTION:

```
# perform both a parametric and nonparametric procedure and compare results
```

Setup for Questions 2 and 3

For the next set of questions, we are going to look at the 20 smallest abalones in terms of Height.

```
cutoff <- sort(abalone$Height)[20]+0.00000001
```

```
# pulling value with tidyverse  
cutoff <- as.numeric(abalone %>%  
  arrange(Height) %>%  
  slice(20) %>%  
  summarize(cutoff = Height + 0.00000001))
```

First, we identify the Height cutoff and add a little bit so we will include the 20th if we ask for heights < the cutoff. In the event of ties, we need to see how many observations we actually end up with.

```
abalone2 <- filter(abalone, Height < cutoff)  
dim(abalone2)
```

```
## [1] 24 9
```

So, we are looking at just 24 abalones now.

Question 2

We tend to presume the smallest abalones are those still developing. In abalone development, how much longer are abalones than they are wide (diameter)?

Remember to think through conditions for applying your chosen statistical procedures.

SOLUTION:

```
# perform an appropriate nonparametric procedure
```

Question 3

A policy maker wants to protect developing abalones (these small ones), and suggests that if the whole weight of a caught abalone is 0.05 or less, that the abalone be released back into the wild to continue developing. Is there evidence to suggest this cutoff is reasonable? If not, what cutoff would you suggest?

Remember to think through conditions for applying your chosen statistical procedures.

SOLUTION:

perform an appropriate nonparametric procedure
note the best procedure may not be the signed rank procedure!

Nonparametric Methods in Literature

As you learn new nonparametric methods, you are likely curious about how they are used in practice. Specifically, how are they used in application areas of interest to you? Let's explore how the signed rank test is used.

Use Google Scholar and do a search for "signed rank test" (or "Wilcoxon signed rank") and an application area of interest to you. For example, search for "signed rank" archaeology' to look for applications of the signed rank test in archaeology. (If the subject area has multiple words, you may want to put quotes around it as well.) Choose an article that pops up from your search that is interesting to you (more recent than 2010 would be best, if possible) and answer the questions below.

If you are having issues picking an application area or choosing an article, some potential articles are provided below. Applications do not need to be in journals in statistics!

Phonlakrai, Monchai, et al., "Assessing Patient and Caregiver Knowledge and Satisfaction on Transcatheter Arterial Chemoembolization Using Video-Based Education," *Journal of Radiology Nursing*, 42.3, 2023: 361-367.

Liu, Longjian, *Heart failure: epidemiology and research methods*, Elsevier Health Sciences, 2017, Chapter 4.

Chandran, Avinash, et al., "Concussion knowledge, attitudes, and self-reporting intentions in youth athletes," *Journal of athletic training*, 55.10, 2020: 1027-1034.

Provide a citation for your article. Google Scholar should be able to provide a complete citation for you to copy/paste.

SOLUTION:

Summarize what the article is about in a few sentences.

SOLUTION:

What was the signed rank test used for in the article? Describe as best you can. For example, what were the hypotheses being tested?

SOLUTION:

Does the article state why the signed rank test was used? If so, what reason do they give?

SOLUTION:

Are any other statistical procedures mentioned in the paper, that you can tell? If so, what else is mentioned?

SOLUTION:

Data Sources

Nash, W., Sellers, T., Talbot, S., Cawthorn, A., & Ford, W. (1994). Abalone [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C55C7W>.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to *Nonparametric Statistical Methods* (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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